

## **BEEE - Unit – IV**

Syllabus: Introduction - Evolution of electronics – Vacuum tubes to nano electronics - Characteristics of PN Junction Diode — Zener Effect — Zener Diode and its Characteristics. Bipolar Junction Transistor — CB, CE, CC Configurations and Characteristics — Elementary Treatment of Small Signal CE Amplifier.

### **Short Questions:**

#### **1. Describe the evolution of electronics.**

**Ans:**

- i. Vacuum Tube Era (1900s): Used bulky vacuum tubes for amplification and switching.
- ii. Transistor Era (1947): Introduction of transistors made devices smaller, reliable, and efficient.
- iii. Integrated Circuit (IC) Era (1960s): Multiple components combined on a single chip, reducing size and cost.
- iv. Microelectronics / VLSI Era (1970s–Present): Development of microprocessors and highly compact circuits with very high speed and functionality.

#### **2. Draw the symbol of Zener Diode.**

**Ans:**



#### **3. Define Zener breakdown voltage.**

**Ans:** Zener breakdown voltage is the minimum reverse voltage at which a Zener diode starts conducting heavily in reverse bias due to the Zener effect. At this voltage, the diode maintains a constant voltage across it even if the current changes.

#### **4. State any two applications of Zener diode.**

**Ans:**

- Voltage Regulation: Used to maintain a constant output voltage in power supplies.
- Voltage Reference: Provides a stable reference voltage in electronic circuits.

**5. Name the three terminals of a transistor.**

**Ans:** The three terminals of a transistor are:

1. Emitter (E)      2. Base (B)      3. Collector (C)

**6. What is the use of Amplifier.**

**Ans:** An amplifier is an electronic device used to increase the strength (amplitude) of a signal without changing its basic shape.

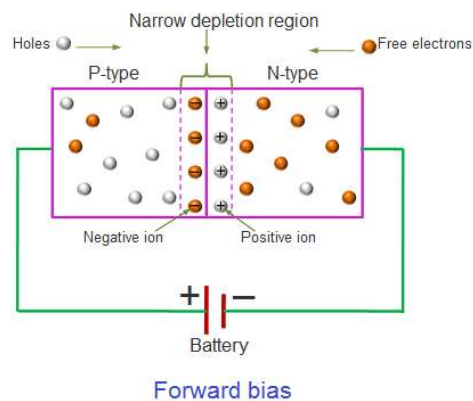
**3. List the three configurations of a Bipolar Junction Transistor (BJT).**

**Ans:**

- a) Common Base (CB) Configuration
- b) Common Emitter (CE) Configuration
- c) Common Collector (CC) Configuration

**7. Draw the circuit-connection of forward-biased pn-junction diode.**

**Ans:**



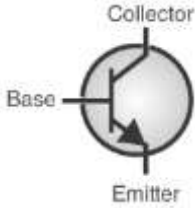
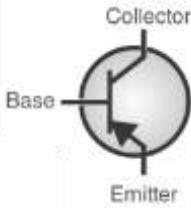
**8. Give two main difference between diode and Zener diode?**

**Ans:**

| Feature   | Diode                                        | Zener Diode                                                 |
|-----------|----------------------------------------------|-------------------------------------------------------------|
| Operation | Conducts only in forward bias                | Conducts in forward bias and reverse bias (after breakdown) |
| Purpose   | Used for rectification (AC to DC conversion) | Used for voltage regulation (maintains constant voltage)    |

**9. What is the difference between NPN and PNP transistor?**

**Ans:**

| Feature               | NPN Transistor                                                                    | PNP Transistor                                                                      |
|-----------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Symbol                |  |  |
| Current Flow          | From collector to emitter                                                         | From emitter to collector                                                           |
| Major Charge Carriers | Electrons                                                                         | Holes                                                                               |

**10. Define voltage gain.**

**Ans:** Voltage gain is the ratio of output voltage to input voltage in an amplifier.

$$A_v = \frac{V_{out}}{V_{in}}$$

It indicates how much the amplifier increases the signal voltage.

**11. What is doping in semi conducting material?**

**Ans:** Doping in semiconductor material is the process of adding a small amount of impurity atoms to a pure semiconductor (like silicon or germanium) to increase its electrical conductivity.

## Long Questions:

### 1. Explain in detail about types of semiconductor materials.

**Ans:** Semiconductors are materials whose electrical conductivity lies between that of conductors and insulators. Their conductivity can be controlled by adding impurities.

Based on purity, semiconductor materials are classified into two types:

1. Intrinsic semiconductors
2. Extrinsic semiconductors.

**Intrinsic Semiconductor:** An intrinsic semiconductor is a pure form of semiconductor material without any impurity atoms. Common examples are silicon and germanium. In these materials, atoms are arranged in a crystal lattice, and each atom shares its valence electrons with neighbouring atoms through covalent bonds.

At absolute zero temperature, intrinsic semiconductors behave like insulators because there are no free charge carriers. However, at room temperature, thermal energy breaks some covalent bonds, generating free electrons and holes. When an electron moves from the valence band to the conduction band, it leaves behind a hole in the valence band. Thus, both electrons and holes act as charge carriers.

In intrinsic semiconductors, the number of electrons is equal to the number of holes, and the conductivity is relatively low. The conductivity increases with temperature because more electron-hole pairs are generated.

**Extrinsic Semiconductor:** An extrinsic semiconductor is obtained by adding a small amount of impurity atoms to a pure semiconductor. This process is called doping. Doping significantly increases the conductivity of the material.

Depending on the type of impurity added, extrinsic semiconductors are classified into two types: N-type and P-type semiconductors.

**N-Type Semiconductor:** An N-type semiconductor is formed by adding a small amount of pentavalent impurity (like Arsenic, Antimony, or Phosphorus) to pure semiconductors (such as Germanium or Silicon). Germanium has 4 valence electrons, while the impurity has 5. Four of the

impurity's electrons form covalent bonds with Germanium atoms, leaving one free electron.

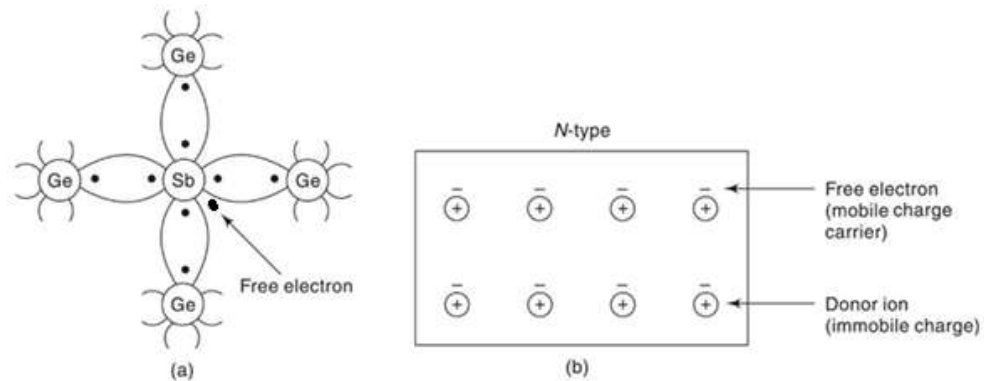


Fig.N-type semiconductor: (a) Formation of covalent bonds (b) Charged carriers

- This free electron can easily move to the conduction band when an electric field or thermal energy is applied, increasing conductivity.
- The impurity is called a donor because it donates an electron for conduction, becoming a positively charged ion fixed in the crystal lattice.
- The result is more free electrons (majority carriers) than holes (minority carriers), boosting conductivity in the N-type semiconductor.

**P-Type Semiconductor:** A P-type semiconductor is formed by adding trivalent impurity (like boron, aluminium, or gallium) to Pure semiconductor material (such as Germanium or Silicon). These atoms have only three valence electrons, so one bond remains incomplete, creating a hole.

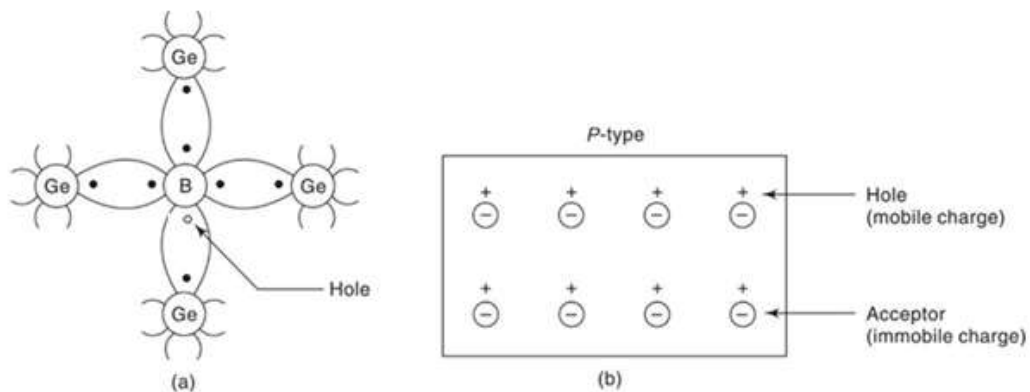


Fig.P-type semiconductor: (a) Formation of covalent bonds (b) Charged carriers

- Boron, with three valence electrons, forms covalent bonds with surrounding Ge atoms, leaving one bond incomplete, creating a hole.
- This increases the number of holes (positive charge carriers), boosting the conductivity of the P-type semiconductor.
- Boron is called an acceptor impurity as it accepts electrons in place of holes. In P-type material, holes are the majority carriers, and electrons are the minority carriers.

## 2. Write the differences between Intrinsic and Extrinsic Semiconductor

**Ans:**

| Aspect                    | Intrinsic Semiconductor                   | Extrinsic Semiconductor                                |
|---------------------------|-------------------------------------------|--------------------------------------------------------|
| Definition                | Pure semiconductor without any impurities | Semiconductor doped with impurities                    |
| Examples                  | Pure Silicon (Si), Germanium (Ge)         | Doped Si or Ge (N-type, P-type)                        |
| Impurity                  | No impurity added                         | Impurities intentionally added (doping)                |
| Charge Carriers           | Equal number of electrons and holes       | Unequal; depends on type (electrons or holes dominate) |
| Conductivity              | Low conductivity                          | Higher conductivity                                    |
| Source of Carriers        | Generated by thermal energy               | Provided by impurity atoms                             |
| Types                     | Only one type                             | Two types: N-type and P-type                           |
| Dependence on Temperature | Strongly dependent                        | Less dependent compared to intrinsic                   |
| Example Dopants           | Not applicable                            | N-type: Phosphorus, Arsenic<br>P-type: Boron, Gallium  |

### 3. Write the differences between N Type and P Type Material.

**Ans:**

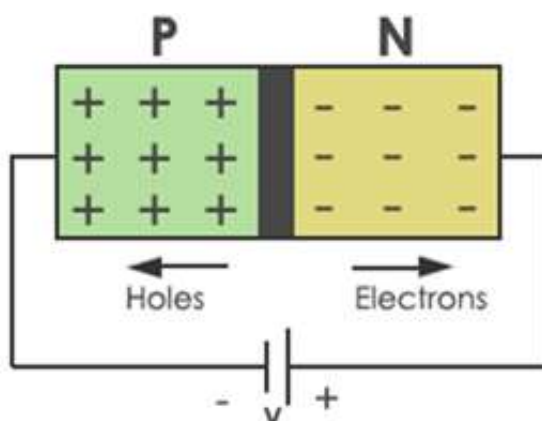
| Aspect                      | N-Type Material                                 | P-Type Material                               |
|-----------------------------|-------------------------------------------------|-----------------------------------------------|
| Definition                  | Semiconductor doped with pentavalent impurities | Semiconductor doped with trivalent impurities |
| Impurity Atoms              | Donor impurities (5 valence electrons)          | Acceptor impurities (3 valence electrons)     |
| Examples of Dopants         | Phosphorus, Arsenic, Antimony                   | Boron, Gallium, Indium                        |
| Majority Carriers           | Electrons                                       | Holes                                         |
| Minority Carriers           | Holes                                           | Electrons                                     |
| Charge of Majority Carriers | Negative (-)                                    | Positive (+)                                  |
| Conductivity                | Due to free electrons                           | Due to holes                                  |
| Symbol Representation       | 'N' indicates negative charge carriers          | 'P' indicates positive charge carriers        |

### 4. Discuss on semiconductor devices and evaluations of electronics?

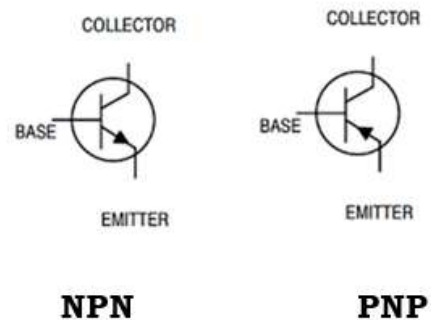
**Ans: Semiconductor Devices:** Semiconductor devices are electronic components made using semiconductor materials such as silicon and germanium. These devices form the foundation of modern electronics and are used to control the flow of current in circuits.

Common Semiconductor Devices:

**1. PN Junction Diode:** A diode is formed by joining P-type and N-type semiconductors. It allows current to flow in one direction (forward bias) and blocks it in the opposite direction (reverse bias). It is used in rectifiers, clippers, and voltage regulators.



**2. Transistor (BJT):** A Bipolar Junction Transistor consists of three layers (emitter, base, collector). It is used for switching and amplification applications. It is of two types: NPN and PNP.



**3. Field Effect Transistor (FET/MOSFET):** A MOSFET is a voltage-controlled semiconductor device used for switching and amplification, where current flow is controlled by the voltage applied to the gate terminal. MOSFETs are widely used in digital circuits due to their high efficiency and low power consumption.

**4. Other Devices:**

- LED (Light Emitting Diode)
- Zener diode (used for voltage regulation)
- Thyristors (SCR, TRIAC) used in power electronics

**Evolution of Electronics:** – Vacuum tubes to nano electronics

The evolution of electronics spans over a century with continuous technological advancements:

1. **1904 – Thermionic Valve:** The first vacuum tube was invented by John Ambrose Fleming, used for rectification.
2. **1906 – Triode:** Lee De Forest developed the triode, capable of signal amplification, widely used in early communication systems.
3. **1947 – Transistor:** John Bardeen, Walter Brattain, and William Shockley invented the transistor, replacing bulky vacuum tubes.
4. **1958 – Integrated Circuits (IC):** Jack Kilby and Robert Noyce developed ICs, integrating multiple components on a single chip.
5. **1971 – Microprocessor:** Intel introduced the first microprocessor (Intel 4004), initiating the microcomputer era.
6. **1980s – VLSI Technology:** Very Large Scale Integration enabled millions of transistors on a single chip, leading to advanced electronic systems.



7. **1990s – Nanotechnology:** Development of nanotechnology allowed manipulation of materials at atomic levels, enabling miniaturization.
8. **1991 – Carbon Nanotubes:** Sumio Iijima discovered carbon nanotubes with excellent electrical and mechanical properties.
9. **2020s – Quantum Computing:** Advances in quantum computing promise high-speed computation using quantum mechanics principles.

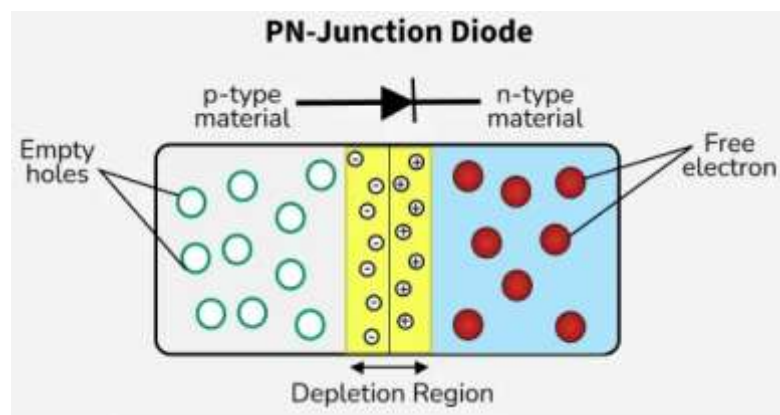
**5. Explain the construction and working of a PN junction diode with neat diagrams. (or) Discuss about the operation of PN junction diode both in Forward bias and Reverse bias conditions with suitable diagrams. (or) Explain PN-junction diode with its characteristics.**

**Ans:** A PN junction diode is a fundamental semiconductor device formed by joining P-type and N-type semiconductor materials. It is widely used in electronic circuits because it allows current to flow only in one direction.

A PN junction diode is formed by joining **P-type** and **N-type** semiconductor materials in a single crystal.

**P-type region:** Doped with trivalent impurities (Boron), contains **holes** as majority carriers

**N-type region:** Doped with pentavalent impurities (Phosphorus), contains **electrons** as majority carriers

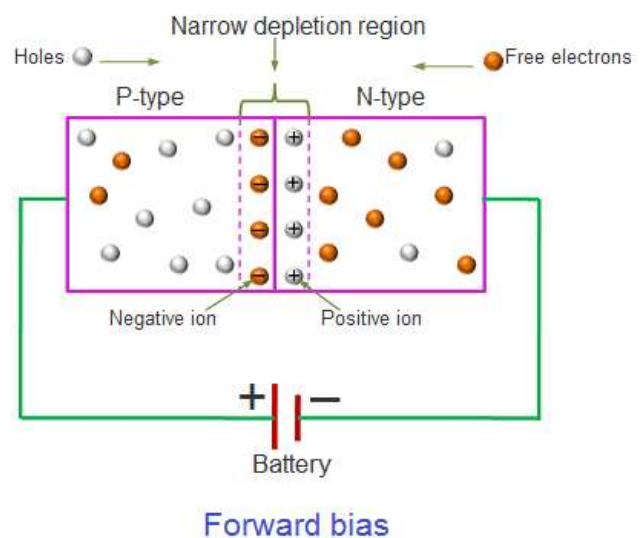


When these two regions are joined together, electrons from the N-side begin to diffuse into the P-side, and holes from the P-side diffuse into the N-side. As a result of this diffusion, electrons and holes recombine near the junction. This recombination leads to the formation of a region that is depleted of free charge carriers, known as the depletion region. Within this depletion

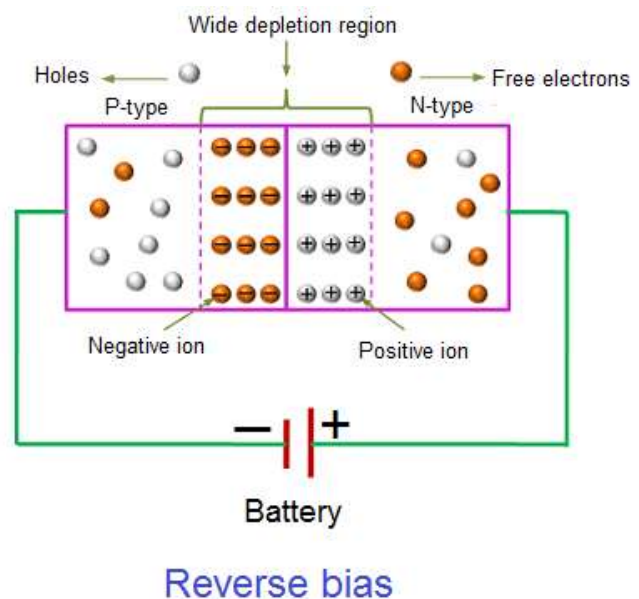
region, immobile ions are left behind, creating an electric field across the junction. This electric field opposes further diffusion of charge carriers and establishes a potential difference called the barrier potential. The barrier potential is approximately 0.7 V for silicon and 0.3 V for germanium.

**Working:** The working of a PN junction diode depends on the external voltage applied across it, which can be either forward bias or reverse bias.

**Forward Bias:** When the diode is connected in forward bias, the P-type region is connected to the positive terminal of the battery and the N-type region to the negative terminal. In this condition, the applied voltage reduces the barrier potential and narrows the depletion region. As the barrier is lowered, charge carriers gain sufficient energy to cross the junction. Electrons from the N-side move toward the P-side, and holes from the P-side move toward the N-side. This movement of charge carriers results in a large current flowing through the diode. Hence, the diode conducts in forward bias.

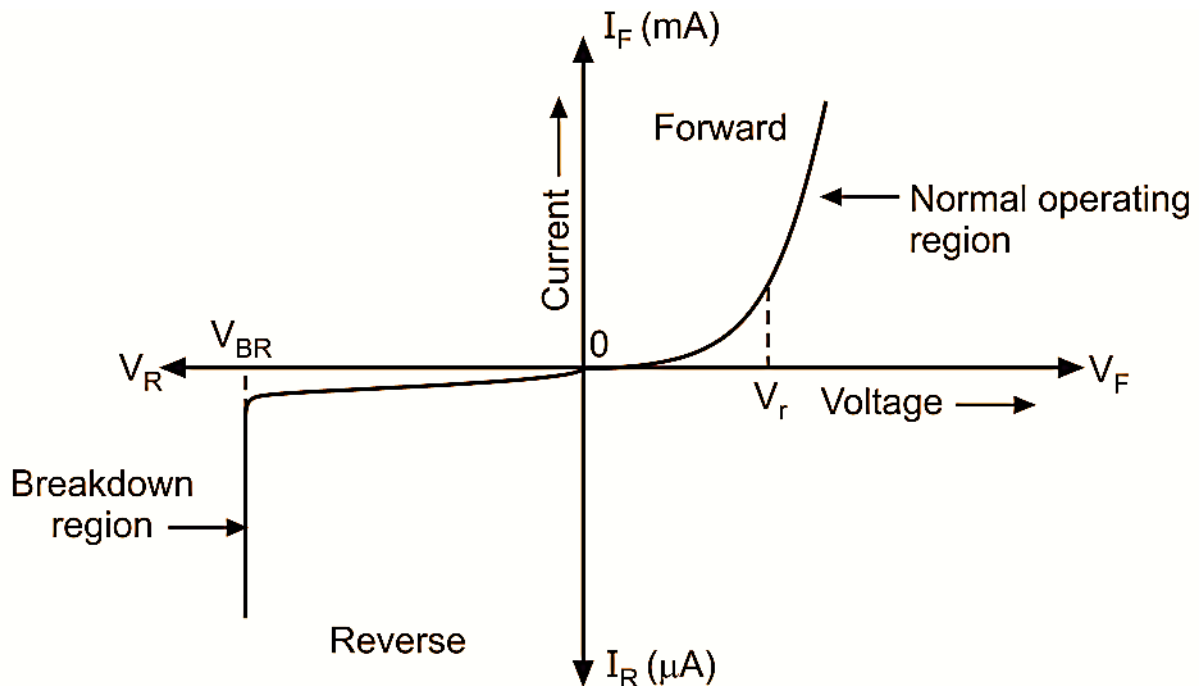


**Reverse Bias:** In reverse bias, the P-type region is connected to the negative terminal and the N-type region to the positive terminal of the battery. In this case, the applied voltage increases the barrier potential and widens the depletion region. The increased barrier prevents the flow of majority charge carriers across the junction. Only a very small current, known as reverse



saturation current, flows due to minority carriers. Therefore, the diode does not conduct in reverse bias.

### V-I Characteristics of PN Junction Diode:



**Fig: V-I Characteristics of PN Junction Diode**

### 6. Explain Zener effect briefly.

Ans: The **Zener effect** is a phenomenon that occurs in a PN junction diode when it is reverse biased and the applied voltage reaches a critical value called the **Zener breakdown voltage**.

“A heavily doped crystal diode which has a sharp breakdown voltage is known as Zener diode”

In a heavily doped PN junction, the depletion region is very thin. When a high reverse voltage is applied, a strong electric field is developed across the junction. This strong field enables electrons to directly move (tunnel) from the valence band of the P-region to the conduction band of the N-region.

As a result, a sudden increase in reverse current occurs without damaging the diode, provided the current is limited. This breakdown is known as the Zener effect.

The Zener effect is mainly observed at **low breakdown voltages** (typically less than 5V) and is widely used in **Zener diodes** for voltage regulation purposes.

**7. Describe and draw the Characteristics of Zener Diode. (or) Describe the operation of a Zener diode in Reverse bias conditions with suitable diagram.**

**Ans:** A Zener diode is a special type of PN junction diode designed to operate in the reverse breakdown region. It is widely used for voltage regulation because it maintains a nearly constant voltage across its terminals over a wide range of current.

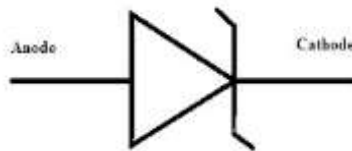


Fig: Symbol of Zener Diode

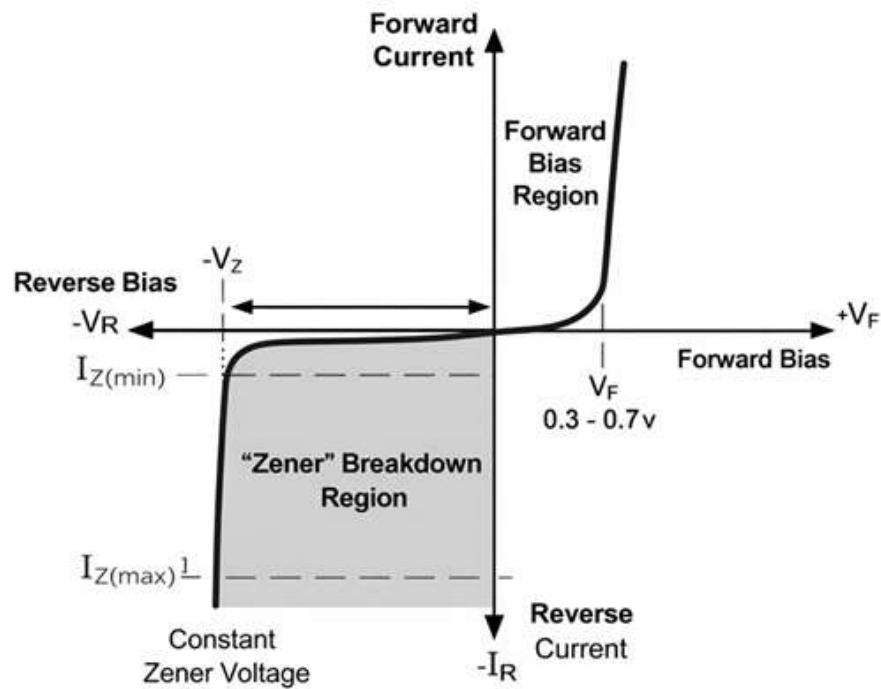
**V-I Characteristics of Zener Diode:** The V-I characteristics of a Zener diode consist of two regions: forward bias and reverse bias.

**Forward Bias Characteristics:** When the Zener diode is forward biased, it behaves like an ordinary PN junction diode. Initially, the current is very small, and once the applied voltage exceeds the threshold value (about 0.7 V for silicon), the current increases rapidly. In this region, the diode offers very low resistance.

**Reverse Bias Characteristics:** When the Zener diode is reverse biased, only a small leakage current flows initially. As the reverse voltage increases and reaches a specific value known as the Zener breakdown voltage ( $V_z$ ), the current increases sharply.

At this point, the voltage across the diode remains almost constant even if the current increases. This is the most important feature of the Zener diode and is used for voltage regulation.

The breakdown occurs due to the Zener effect (in heavily doped diodes) or avalanche effect (in lightly doped diodes).



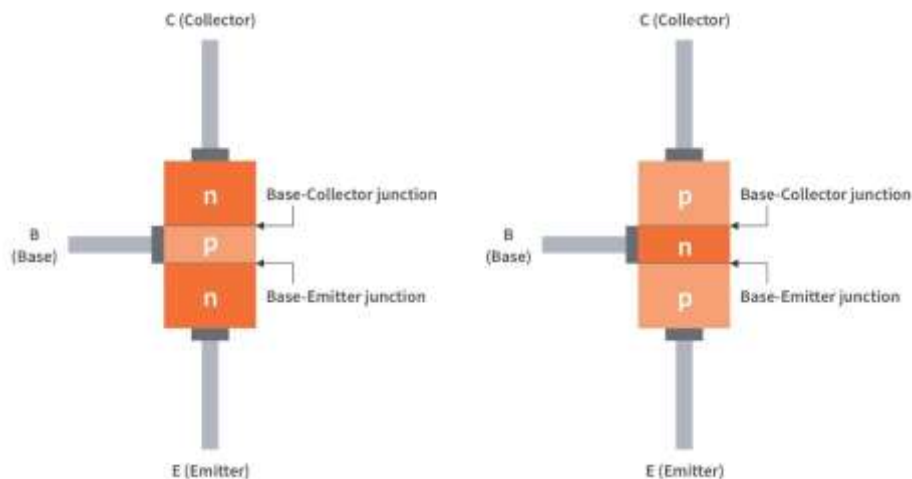
**Fig: V-I Characteristics of Zener Diode**

**8. Explain about working of BJT Transistor.**

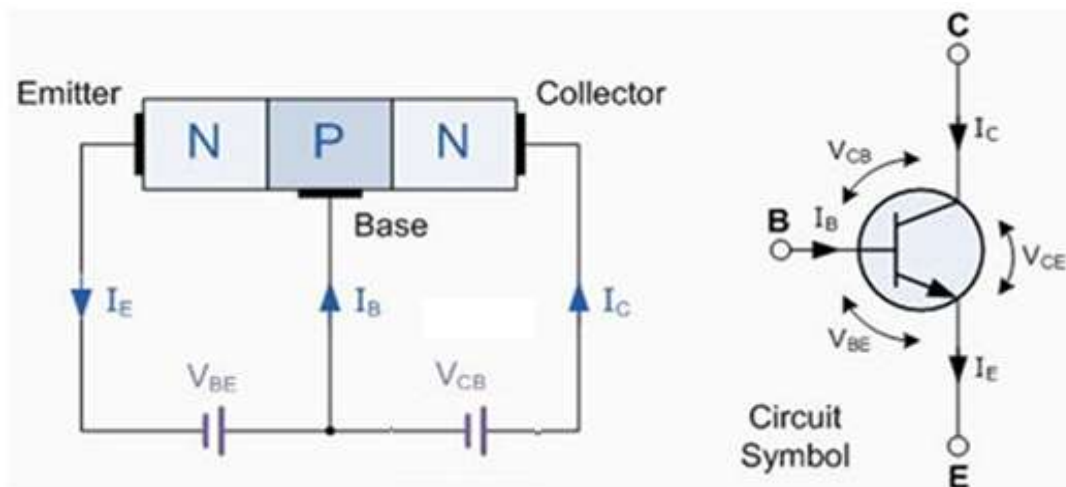
**Ans:**

A Bipolar Junction Transistor (BJT) is a three-layer semiconductor device consisting of two p-n junctions. It has three terminals namely emitter, base, and collector.

The emitter is heavily doped to supply charge carriers, the base is very thin and lightly doped, and the collector is moderately doped to collect carriers. A BJT can be of two types: NPN and PNP, but the working principle remains similar.



The operation of a BJT is mainly explained using the NPN transistor, which is widely used. For proper operation in the amplification mode (active region), the emitter-base junction is forward biased and the collector-base junction is reverse biased. When the emitter-base junction is forward biased, electrons are injected from the emitter into the base region. Since the base is very thin and lightly doped, only a small number of electrons recombine with holes in the base, producing a small base current. The majority of electrons pass through the base and reach the collector-base junction.



At the collector-base junction, which is reverse biased, an electric field exists that attracts and pulls these electrons into the collector. Thus, a large collector current flows even though the base current is very small. This is the fundamental principle of transistor action: a small base current controls a large collector current.

The currents in the transistor are related as:

$$I_E = I_B + I_C$$

and the collector current is approximately proportional to the base current:

$$I_C = \beta I_B$$

where  $\beta$  is the current gain of the transistor.

The transistor operates in three regions. In the cut-off region, both junctions are reverse biased and no current flows, so the transistor acts as an open switch. In the active region, the emitter-base junction is forward biased and the collector-base junction is reverse biased, allowing

amplification of signals. In the saturation region, both junctions are forward biased and maximum current flows, so the transistor acts as a closed switch.

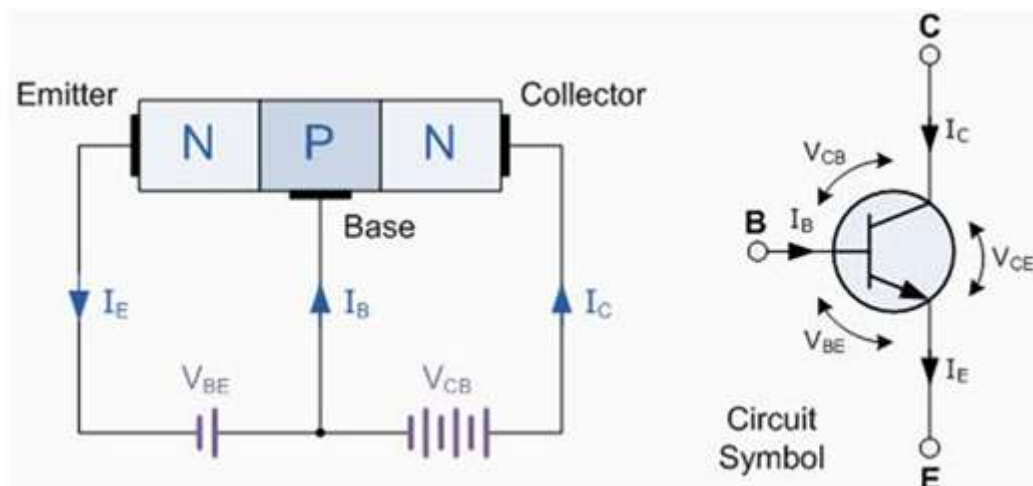
In summary, the BJT is a current-controlled device in which the conduction depends on both electrons and holes. Its ability to control a large current using a small input current makes it very useful in amplifiers and switching circuits.

### 9. Discuss in detail about PNP and NPN transistors.

Ans: A transistor is a semiconductor device used to amplify signals and switch electronic circuits. It is one of the most important components in modern electronics.

**NPN Transistor:** An NPN transistor is formed by sandwiching a P-type semiconductor between two N-type semiconductor regions. In this type, electrons are the majority charge carriers.

In operation, the emitter-base junction is forward biased and the collector-base junction is reverse biased. When a small current flows into the base, a large number of electrons move from the emitter to the collector. This results in a large collector current. Thus, a small base current controls a large collector current, making the transistor useful for amplification.

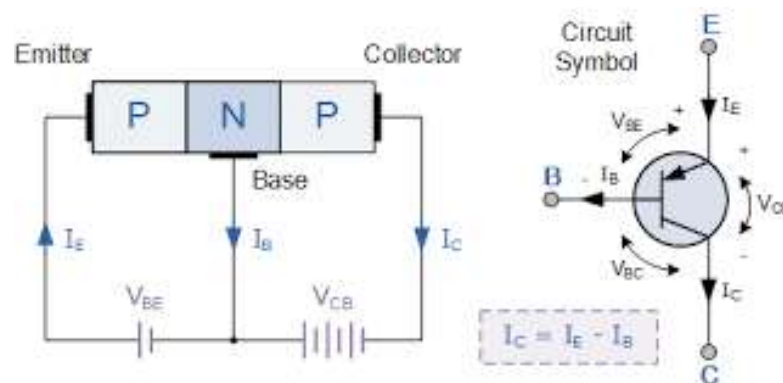


The current flow in an NPN transistor is mainly due to the movement of electrons, which gives it better performance and faster operation compared to PNP transistors. In the symbol of an NPN transistor, the arrow on the emitter points outward, indicating the direction of conventional current.

**PNP Transistor:** A PNP transistor is formed by sandwiching an N-type semiconductor between two P-type semiconductor regions. In this type, holes are the majority charge carriers.

In operation, the emitter-base junction is forward biased and the collector-base junction is reverse biased, similar to the NPN transistor but with opposite polarity. When a small base current flows out of the base, holes move from the emitter to the collector, producing the collector current.

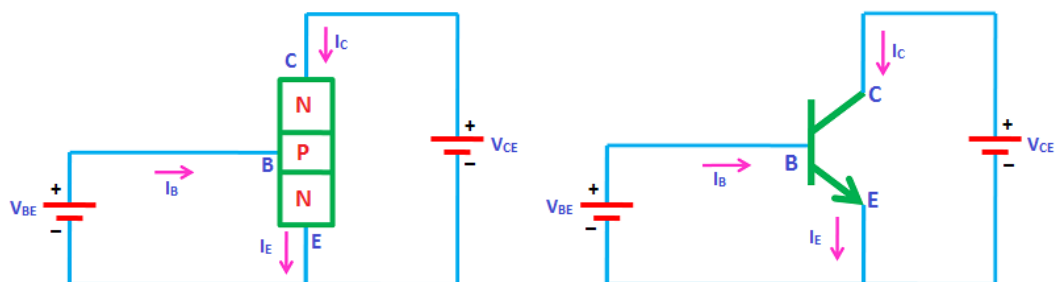
The current flow in a PNP transistor is mainly due to holes. In the symbol of a PNP transistor, the arrow on the emitter points inward, indicating the direction of conventional current flow.



# 10. Describe the characteristics of the Common Emitter (CE) configuration in a Bipolar Junction Transistor (BJT)?

Ans: **Characteristics of Common Emitter (CE) Configuration in BJT:**

In the **Common Emitter (CE) configuration**, the emitter terminal of a Bipolar Junction Transistor (BJT) is common to both the input and output circuits.



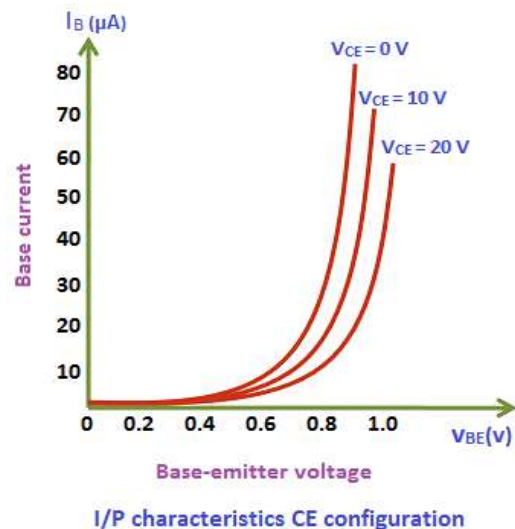
Common emitter configuration



The input is applied between the base and emitter, and the output is taken between the collector and emitter. This configuration is widely used because it provides high current gain, voltage gain, and power gain.

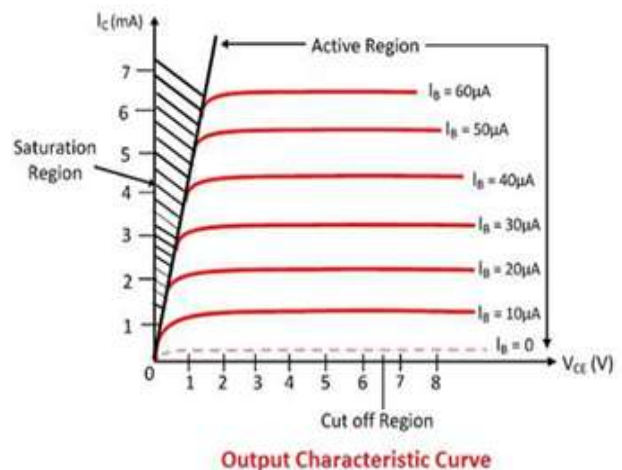
**Input Characteristics:** The input characteristic of a CE configuration represents the relationship between the base current ( $I_B$ ) and the base-emitter voltage ( $V_{BE}$ ) at a constant collector-emitter voltage ( $V_{CE}$ ).

Initially, when  $V_{BE}$  is small, the base current is very low. Once the applied voltage exceeds the cut-in voltage (about 0.7 V for silicon), the base current increases rapidly. The curve resembles that of a forward-biased diode because the base-emitter junction is forward biased.



**Output Characteristics:** The output characteristic shows the relationship between the collector current ( $I_C$ ) and the collector-emitter voltage ( $V_{CE}$ ) for different values of base current ( $I_B$ ).

At low  $V_{CE}$ , the transistor operates in the **saturation region**, where both junctions are forward biased and the collector current increases rapidly. In the middle region, known as the **active region**, the collector current remains almost constant for a given base current and is mainly controlled by  $I_B$ . This region is used for amplification.



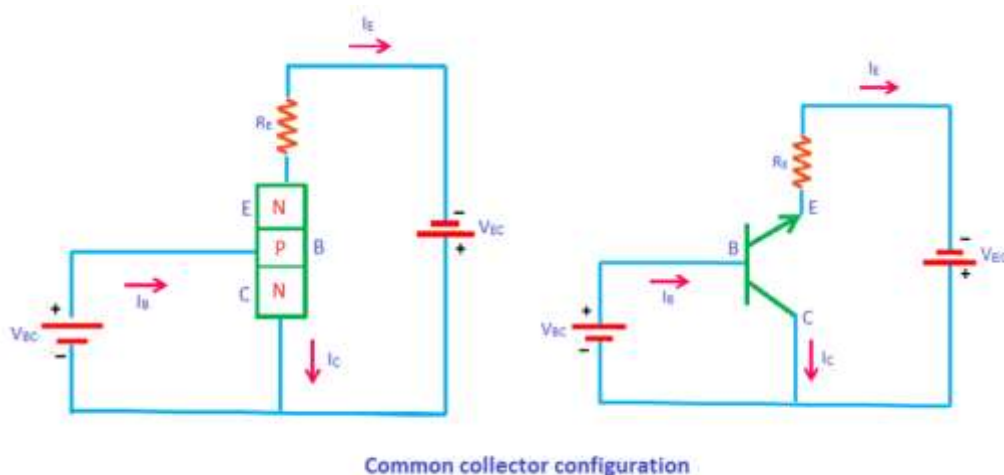
At very high  $V_{CE}$ , the transistor enters the **cut-off region**, where the base current is nearly zero and the collector current is also very small.

**Advantages of CE Configuration:** The CE configuration is widely used because it provides high current gain, high voltage gain, and good power gain. It is suitable for amplification purposes in most electronic circuits.

11. **Explain about CC configuration.**

**Ans: Common Collector (CC) Configuration of BJT:** In the Common Collector (CC) configuration, the collector terminal of the transistor is common to both the input and output circuits. The input is applied between the base and collector, while the output is taken between the emitter and collector. This configuration is also known as the **emitter follower**.

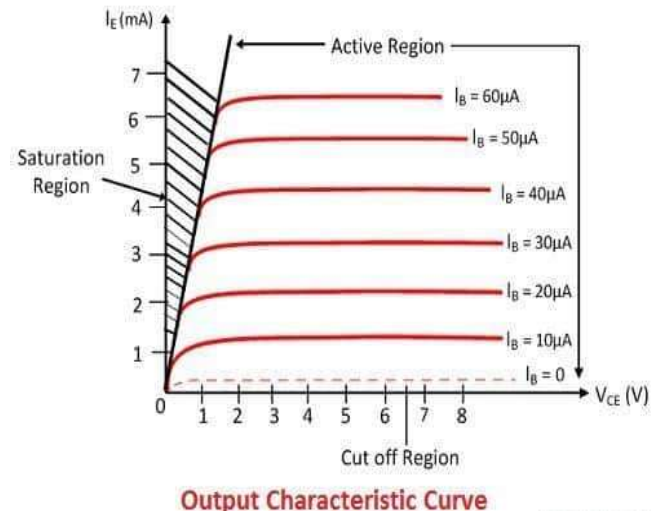
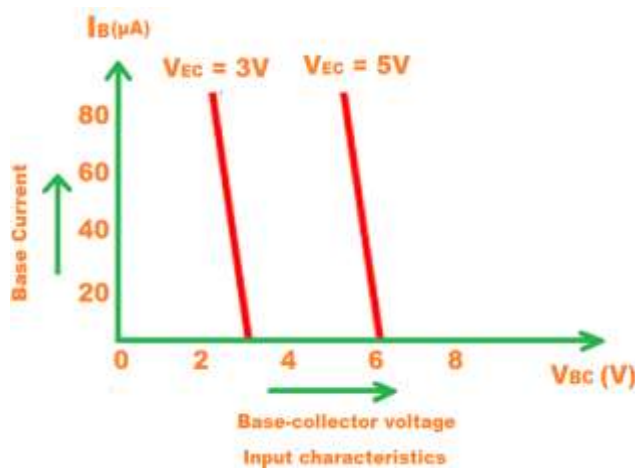
**Working Principle:** In this configuration, the base-emitter junction is forward biased and the collector-base junction is reverse biased. When an input voltage is applied to the base, it controls the emitter current.



The output voltage is taken across the emitter and is nearly equal to the input voltage, but slightly less due to the base-emitter voltage drop (about 0.7 V for silicon). Hence, the output follows the input, which is why it is called an emitter follower.

**Input Characteristics:** The input characteristic shows the relationship between base current ( $I_B$ ) and base-collector voltage. It is similar to a forward-biased diode, as the base-emitter junction is forward biased.

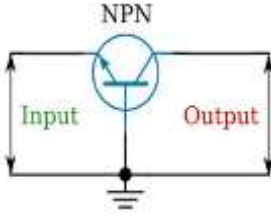
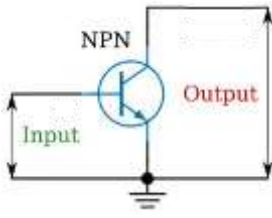
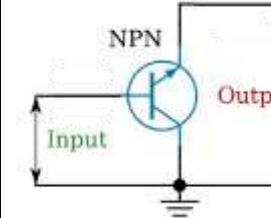
**Output Characteristics:** The output characteristic shows the relationship between emitter current ( $I_E$ ) and emitter-collector voltage for constant base current. The emitter current increases with increase in base current.



## 12. Differentiate between BJT CB, CE, CC Configurations.

Ans:

| Parameter                | CB<br>(Common Base)                  | CE<br>(Common Emitter)    | CC<br>(Common Collector/ Emitter Follower) |
|--------------------------|--------------------------------------|---------------------------|--------------------------------------------|
| Circuit Diagram          |                                      |                           |                                            |
| Input Terminal           | Emitter                              | Base                      | Base                                       |
| Output Terminal          | Collector                            | Collector                 | Emitter                                    |
| Common Terminal          | Base                                 | Emitter                   | Collector                                  |
| Input Resistance         | Low                                  | Medium                    | High                                       |
| Output Resistance        | High                                 | Medium                    | Low                                        |
| Current Gain ( $\beta$ ) | $< 1$ ( $\alpha \approx 0.95-0.99$ ) | High ( $\approx 20-200$ ) | Very High ( $\approx \beta+1$ )            |
| Voltage Gain             | High                                 | High                      | $\approx 1$ (Unity)                        |
| Power Gain               | Medium                               | Very High                 | High                                       |

| Parameter       | CB<br>(Common Base)                                                               | CE<br>(Common Emitter)                                                             | CC<br>(Common Collector/ Emitter Follower)                                          |
|-----------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Circuit Diagram |  |  |  |
| Phase Shift     | No phase reversal                                                                 | 180° phase reversal                                                                | No phase reversal                                                                   |
| Bandwidth       | High                                                                              | Medium                                                                             | High                                                                                |
| Application     | High-frequency amplifiers                                                         | General amplifiers                                                                 | Buffer / impedance matching                                                         |

**13. Write a short note on Small Signal CE amplifier.**

**Ans:** A Small Signal Common Emitter (CE) Amplifier is a single-stage transistor amplifier that uses a Bipolar Junction Transistor (BJT) to increase the amplitude of a weak input signal (typically in millivolts) without significant distortion. The "common emitter" configuration refers to the emitter terminal being shared between the input and output circuits, usually grounded for AC signals, offering high voltage and power gain.

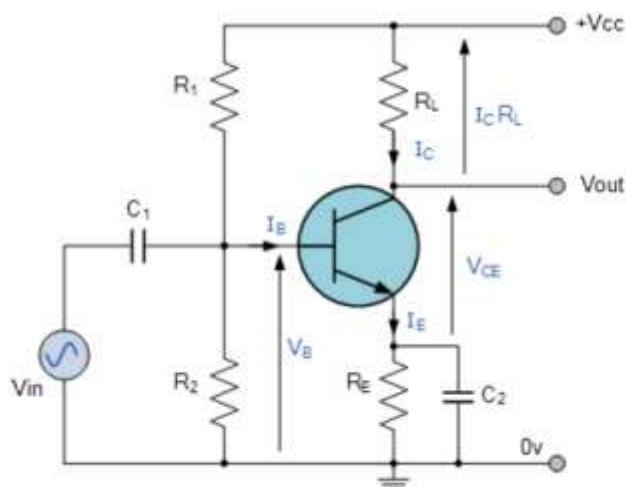


Fig: Circuit diagram

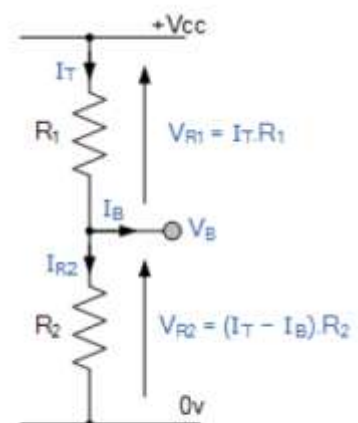


Fig. Voltage divider biasing

The single stage common emitter amplifier circuit shown above uses what is commonly called “**Voltage Divider Biasing**”. This type of biasing arrangement uses two resistors as a potential divider network across the supply with their centre point supplying the required Base bias voltage to the transistor. Voltage divider biasing is commonly used in the design of bipolar transistor amplifier circuits.

This method of biasing the transistor greatly reduces the effects of varying Beta, ( $\beta$ ) by holding the Base bias at a constant steady voltage level allowing for best stability.

The quiescent Base voltage ( $V_B$ ) is determined by the potential divider network formed by the two resistors,  $R_1$ ,  $R_2$ , and the power supply voltage  $V_{CC}$  as shown with the current flowing through both resistors. Then the total resistance  $R_T$  will be equal to  $R_1 + R_2$  giving the current as  $i = V_{CC}/R_T$ . The voltage level generated at the junction of resistors  $R_1$  and  $R_2$  holds the Base voltage ( $V_b$ ) constant at a value below the supply voltage. The potential divider network used in the common emitter amplifier circuit divides the supply voltage in proportion to the resistance.

This bias reference voltage can be easily calculated using the simple voltage divider formula below:

**Transistor Bias Voltage,**

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2}$$

As the same supply voltage, ( $V_{CC}$ ) also determines the maximum Collector current,  $I_c$  when the transistor is switched fully “ON” (saturation),  $V_{CE} = 0$ . The Base current  $I_B$  for the transistor is found from the Collector current,  $I_c$  and the DC current gain Beta,  $\beta$  of the transistor.

Beta Value,

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

A transistor’s Beta value, sometimes referred to as  $h_{FE}$  on datasheets, defines the transistor’s forward current gain in the common emitter

configuration. Beta is an electrical parameter built into the transistor during manufacture. Beta ( $h_{FE}$ ) has no units as it is a fixed ratio of the two currents,  $I_C$  and  $I_B$  so a small change in the Base current will cause a large change in the Collector current.